

Course Map

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1. Overview

Key point: The Course map ties every Note together. It shows where each idea sits in an ML project, how ideas connect, what to read first, and which algorithm to choose.

The Notes each teach one lesson. The Course map shows how those lessons fit together, in four views:

View	Question it answers
Pipeline map	Where does this idea sit in a real ML project?
Concept map	How is this idea connected to the others?
Learning path	Which Notes should I read first?
Algorithm chooser	Which algorithm suits my problem?

Each idea on the map is a **Concept**. Concepts already taught in a written Note are **confirmed**; the others are **draft**, shown faint, and are checked against their Video when their Note is written. So far, 126 of 149 Concepts are confirmed.

Every Note starts with a *Where this fits* box: a small Pipeline map with that Note's steps highlighted, plus what it builds on and what it leads to.

Extra: This map is generated from one data file, `course_map/concepts.yaml`. An interactive version, where you can zoom, filter by step and click a Concept to see its links and Notes, runs with `python course_map/app.py`.

2. The Pipeline map

Key point: Every ML project moves through the same steps, from framing the problem to monitoring the deployed model. Every Concept belongs to one step.

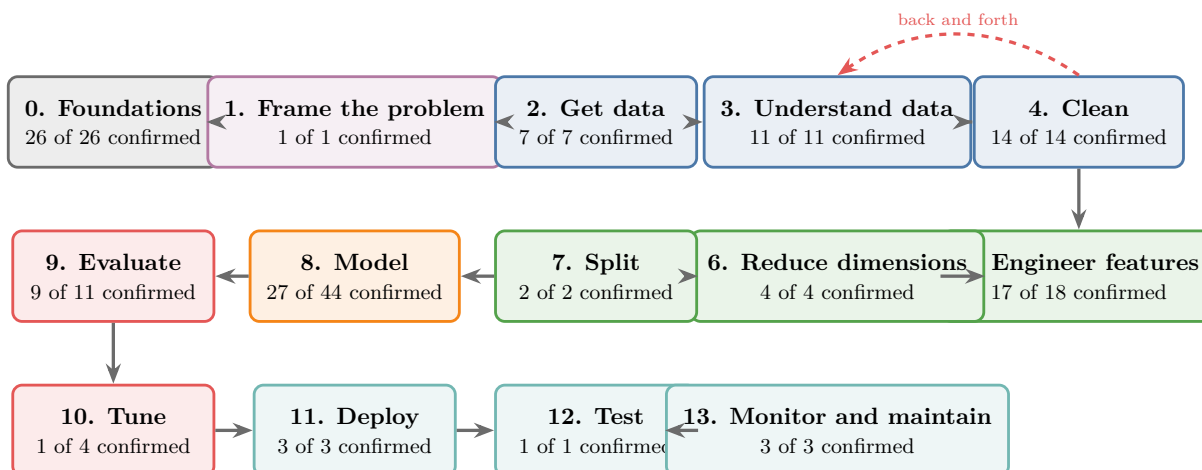


Figure 1: The Pipeline map: 14 steps of an ML project

Figure 1 shows the 14 steps. They follow the ML development life cycle: frame the problem, get and understand the data, clean it, engineer features, train and evaluate models, tune them, deploy, test and keep the model healthy.

Steps 3 (Understand data) and 4 (Clean) loop: exploring reveals what needs cleaning, and cleaning changes the data, so we explore again.

2.1 Step 0: Foundations

Concept	Taught in	Status
Machine learning	Note 1 , Note 2	confirmed
Data mining	Note 1 , Note 8	confirmed
Artificial intelligence	Note 2	confirmed
Symbolic AI and expert systems	Note 2	confirmed
Deep learning	Note 2	confirmed
Neural networks	Note 2	confirmed
Features	Note 2 , Note 11	confirmed
Supervised learning	Note 3	confirmed
Regression problems	Note 3	confirmed
Classification problems	Note 3	confirmed
Unsupervised learning	Note 3	confirmed
Semi-supervised learning	Note 3	confirmed
Reinforcement learning	Note 3	confirmed
Batch (offline) learning	Note 4	confirmed
Online learning	Note 5	confirmed
Out-of-core learning	Note 5	confirmed
Instance-based learning	Note 6	confirmed
Model-based learning	Note 6	confirmed
Applications of ML	Note 8	confirmed
ML development life cycle	Note 9 , Note 13	confirmed
Tensors	Note 11	confirmed
Setup: conda, Jupyter and Colab	Note 12	confirmed
Eigenvectors and eigenvalues	Note 48	confirmed
Conditional probability	Note 82	confirmed
Independent and mutually exclusive events	Note 83 , Note 84	confirmed
Bayes' theorem	Note 85 , Note 86	confirmed

2.2 Step 1: Frame the problem

Concept	Taught in	Status
Framing an ML problem	Note 9 , Note 14	confirmed

2.3 Step 2: Get data

Concept	Taught in	Status
Labelled data	Note 3 , Note 7	confirmed
Enough data	Note 7	confirmed
Sampling noise and bias	Note 7	confirmed
APIs	Note 7 , Note 9 , Note 17	confirmed
Web scraping	Note 7 , Note 9 , Note 18	confirmed
CSV files	Note 9 , Note 13 , Note 15	confirmed
JSON and SQL data	Note 16	confirmed

2.4 Step 3: Understand data

Concept	Taught in	Status
Exploratory data analysis	Note 9 , Note 13 , Note 19	confirmed
Univariate analysis	Note 9 , Note 20	confirmed
Bivariate and multivariate analysis	Note 9 , Note 21	confirmed
Imbalanced data	Note 9 , Video 133, coming	confirmed
Variance	Note 19 , Note 47	confirmed
Correlation	Note 19 , Note 21	confirmed
Descriptive statistics	Note 19	confirmed
Skewness	Note 20	confirmed
Pandas Profiling	Note 22	confirmed
Q-Q plot	Note 30	confirmed
Covariance and covariance matrix	Note 48	confirmed

2.5 Step 4: Clean

Concept	Taught in	Status
Poor-quality data	Note 7 , Note 9	confirmed
Missing values	Note 7 , Note 9 , Note 23 , Note 35 , Note 36 , Note 37 , Note 38	confirmed
Outliers	Note 7 , Note 9 , Note 20 , Note 23 , Note 41	confirmed

Concept	Taught in	Status
Simple imputation (mean, median, mode, constant)	Note 23 , Note 36 , Note 37	confirmed
Complete case analysis	Note 35	confirmed
Missing indicator	Note 38	confirmed
Random sample imputation	Note 38	confirmed
KNN imputer	Note 39	confirmed
Iterative imputation (MICE)	Note 40	confirmed
Trimming outliers	Note 41 , Note 42 , Note 43 , Note 44	confirmed
Capping (winsorization)	Note 41 , Note 42 , Note 43 , Note 44	confirmed
Z-score outlier method	Note 41 , Note 42	confirmed
IQR outlier method	Note 41 , Note 43	confirmed
Percentile outlier method	Note 41 , Note 44	confirmed

2.6 Step 5: Engineer features

Concept	Taught in	Status
Feature scaling	Note 6 , Note 9 , Note 13 , Note 23 , Note 24	confirmed
Feature engineering	Note 7 , Note 9 , Note 23	confirmed
Feature construction and splitting	Note 7 , Note 9 , Note 23 , Note 45	confirmed
Feature selection	Note 9 , Note 13 , Note 23 , Note 46	confirmed
Standardization	Note 9 , Note 13 , Note 24	confirmed
One-hot encoding	Note 11 , Note 23 , Note 27	confirmed
ML pipelines	Note 13 , Note 29 , Note 38	confirmed
Encoding categorical data	Note 23 , Note 26	confirmed
Binning and binarization	Note 23 , Note 32	confirmed
Feature transformation	Note 23	confirmed
Normalization	Note 25	confirmed
Ordinal and label encoding	Note 26	confirmed
Column transformer	Note 28	confirmed
Function transformer	Note 30	confirmed
Power transformer	Note 31	confirmed
Mixed variables	Note 33	confirmed
Date and time features	Note 34	confirmed
Feature importance	Video 114, coming	draft

2.7 Step 6: Reduce dimensions

Concept	Taught in	Status
Dimensionality reduction	Note 3 , Note 46	confirmed
PCA	Note 3 , Note 47 , Note 48 , Note 49	confirmed
Feature extraction	Note 23 , Note 46 , Note 47	confirmed
Curse of dimensionality	Note 46 , Note 91	confirmed

2.8 Step 7: Split

Concept	Taught in	Status
Train-test split	Note 13	confirmed
Data leakage	Note 13 , Note 91	confirmed

2.9 Step 8: Model

Concept	Taught in	Status
Clustering	Note 3 , Video 128, coming	confirmed
Anomaly detection	Note 3	confirmed
Association rule learning	Note 3	confirmed
Stochastic gradient descent	Note 5 , Note 59	confirmed
K-nearest neighbours	Note 6 , Note 91	confirmed
Ensemble learning	Note 9 , Video 101, coming	draft
Logistic regression	Note 13 , Note 70 , Note 71 , Note 72 , Note 73 , Note 75	draft
Multicollinearity	Note 27	confirmed
Simple linear regression	Note 50 , Note 51	confirmed
Best-fit line and squared error	Note 50 , Note 51	confirmed
Ordinary least squares (closed form)	Note 51	confirmed
Multiple linear regression	Note 53 , Note 54 , Note 55	confirmed
Normal equation	Note 54 , Note 55	confirmed
Assumptions of linear regression	Note 56	confirmed
Gradient descent	Note 57	confirmed
Convex and non-convex loss	Note 57	confirmed
Batch gradient descent	Note 58	confirmed
Mini-batch gradient descent	Note 60	confirmed
Polynomial regression	Note 61	confirmed
Polynomial features	Note 61 , Note 80	confirmed
Regularisation	Note 63	confirmed

Concept	Taught in	Status
Ridge regression	Note 63 , Note 64 , Note 65 , Note 66	confirmed
Lasso regression	Note 67 , Note 68	confirmed
Elastic Net	Note 69	confirmed
Perceptron trick	Note 70 , Note 71	confirmed
Sigmoid function	Note 72 , Note 74	confirmed
Log loss (binary cross entropy)	Note 73	confirmed
Softmax regression	Note 79	confirmed
Naive Bayes	Note 87 , Note 88 , Note 89 , Note 90	confirmed
Support vector machines	Video 92, coming, Video 93, coming, Video 94, coming	draft
Kernel trick	Video 95, coming, Video 96, coming	draft
Decision trees	Video 97, coming, Video 98, coming, Video 100, coming	draft
Regression trees	Video 99, coming	draft
Voting ensembles	Video 102, coming, Video 103, coming, Video 104, coming	draft
Bagging	Video 105, coming, Video 106, coming, Video 107, coming	draft
Random forest	Video 108, coming, Video 109, coming, Video 110, coming, Video 111, coming	draft
AdaBoost	Video 115, coming, Video 116, coming, Video 117, coming, Video 118, coming	draft
Boosting	Video 119, coming	draft
Gradient boosting	Video 120, coming, Video 121, coming, Video 122, coming	draft
XGBoost	Video 123, coming, Video 124, coming, Video 125, coming, Video 126, coming	draft
Stacking and blending	Video 127, coming	draft
K-means	Video 128, coming, Video 129, coming, Video 130, coming	draft
Hierarchical clustering	Video 131, coming	draft

DBSCAN	Video 132, coming	draft
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2.10 Step 9: Evaluate

Concept	Taught in	Status
Overfitting	Note 7 , Note 61 , Note 91	confirmed
Underfitting	Note 7 , Note 61 , Note 91	confirmed
Accuracy	Note 13 , Note 76 , Note 91	confirmed
Cross-validation	Note 29 , Note 30 , Note 91 , Video 112, coming	draft
Regression metrics	Note 52	confirmed
Bias-variance trade-off	Note 62 , Video 109, coming	confirmed
Confusion matrix	Note 76	confirmed
Precision, recall and F1	Note 77	confirmed
ROC curve and AUC	Note 78	confirmed
Decision surface and boundary	Note 91	confirmed
OOB score	Video 113, coming	draft

2.11 Step 10: Tune

Concept	Taught in	Status
Hyperparameter tuning	Note 9 , Note 81 , Note 91 , Video 98, coming, Video 111, coming, Video 118, coming	draft
Grid and random search	Note 29 , Note 38 , Video 112, coming	draft
Learning rate	Note 57	confirmed
Optuna	Video 134, coming	draft

2.12 Step 11: Deploy

Concept	Taught in	Status
Deployment	Note 4 , Note 7 , Note 9 , Note 13 , Note 29	confirmed
Software integration	Note 7	confirmed
Saving models with pickle	Note 9 , Note 13 , Note 29	confirmed

2.13 Step 12: Test

Concept	Taught in	Status
Beta and A/B testing	Note 9	confirmed

2.14 Step 13: Monitor and maintain

Concept	Taught in	Status
Model drift	Note 4 , Note 9	confirmed
Retraining	Note 4 , Note 5 , Note 9	confirmed
MLOps and cost	Note 7 , Note 9	confirmed

3. The Concept map

Key point: Concepts are joined by five kinds of Link. Following the Links shows why each idea exists and what it is for.

Every Link has one of five types:

Link	Meaning	Example
needs	must be understood first	logistic regression <i>needs</i> gradient descent
is a kind of	a special case	Ridge <i>is a kind of</i> regularisation
fixes	solves a problem	regularisation <i>fixes</i> overfitting
compared with	often confused or contrasted	bagging <i>compared with</i> boosting
used in	a tool used inside something else	feature scaling <i>used in</i> KNN

The full map has 149 Concepts, too many for one page, so it is shown in six areas (Figures 2 to 7). In each figure, the area's own Concepts are large and coloured by pipeline step; Concepts from other areas that link in are small and grey. Faint dots and dotted lines are drafts.

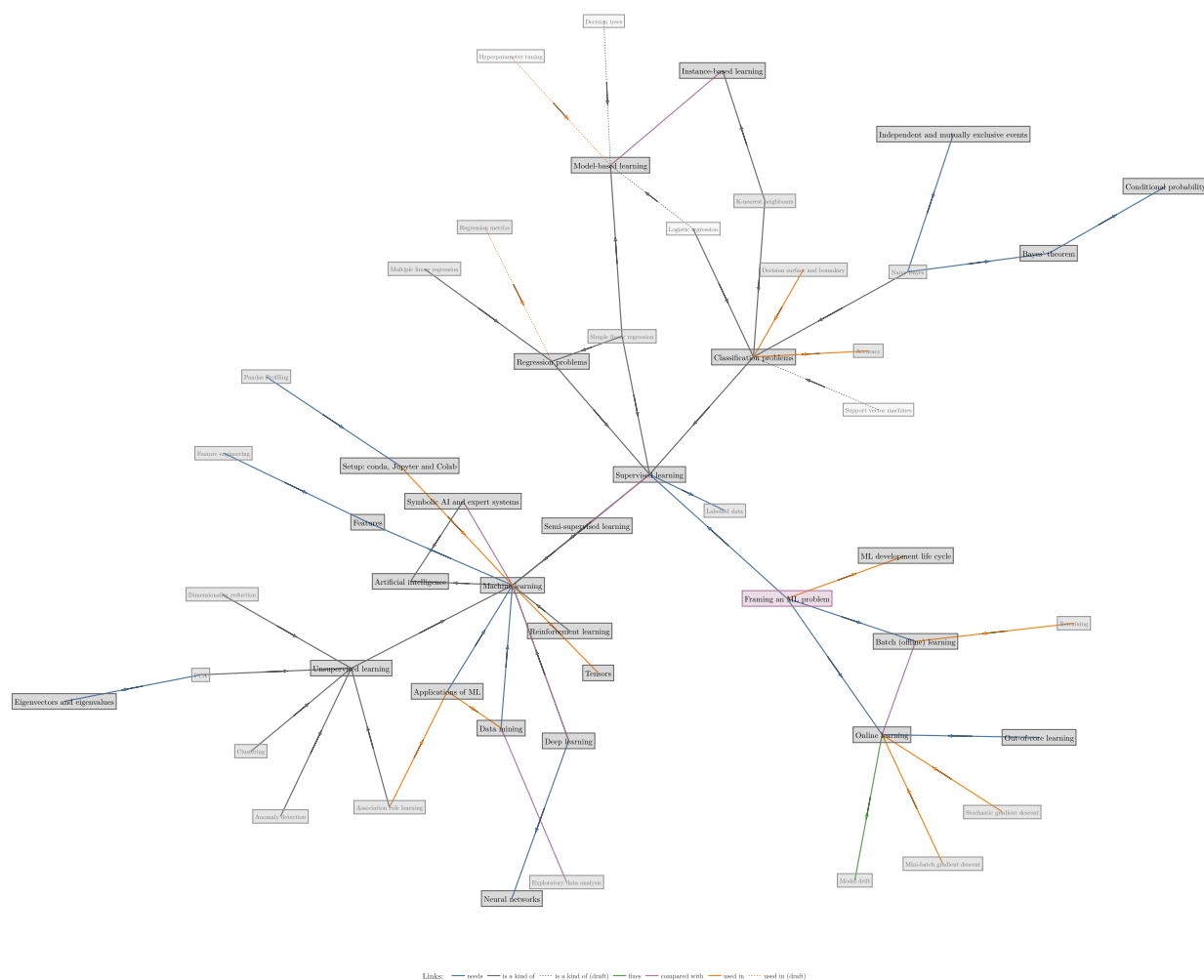


Figure 2: Concept map: Foundations and framing

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Concept map: Features, dimensions and splitting

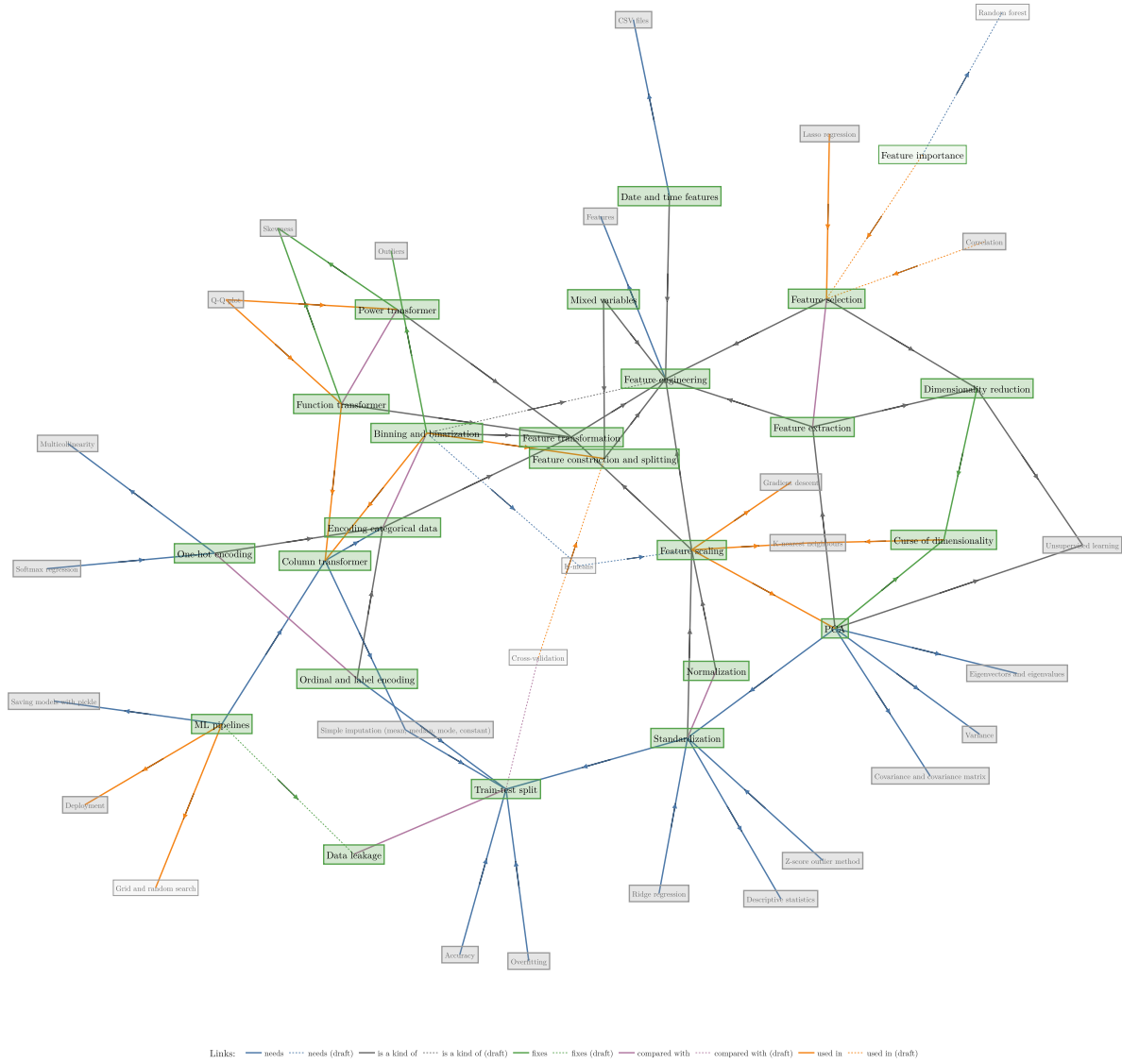


Figure 4: Concept map: Features, dimensions and splitting

Concept map: Models: regression, classification and gradient descent

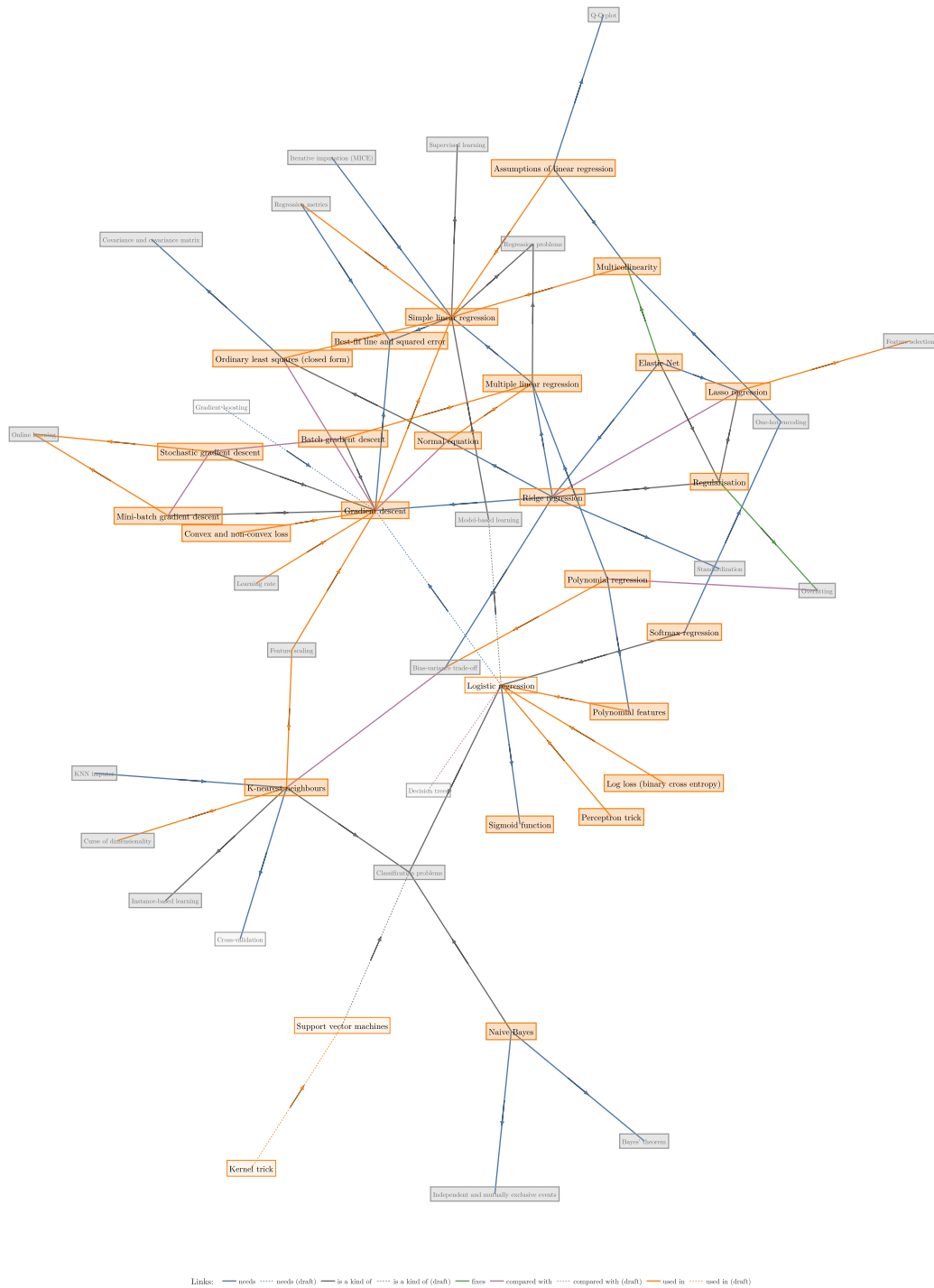


Figure 5: Concept map: Models: regression, classification and gradient descent

Concept map: Models: trees, ensembles and clustering

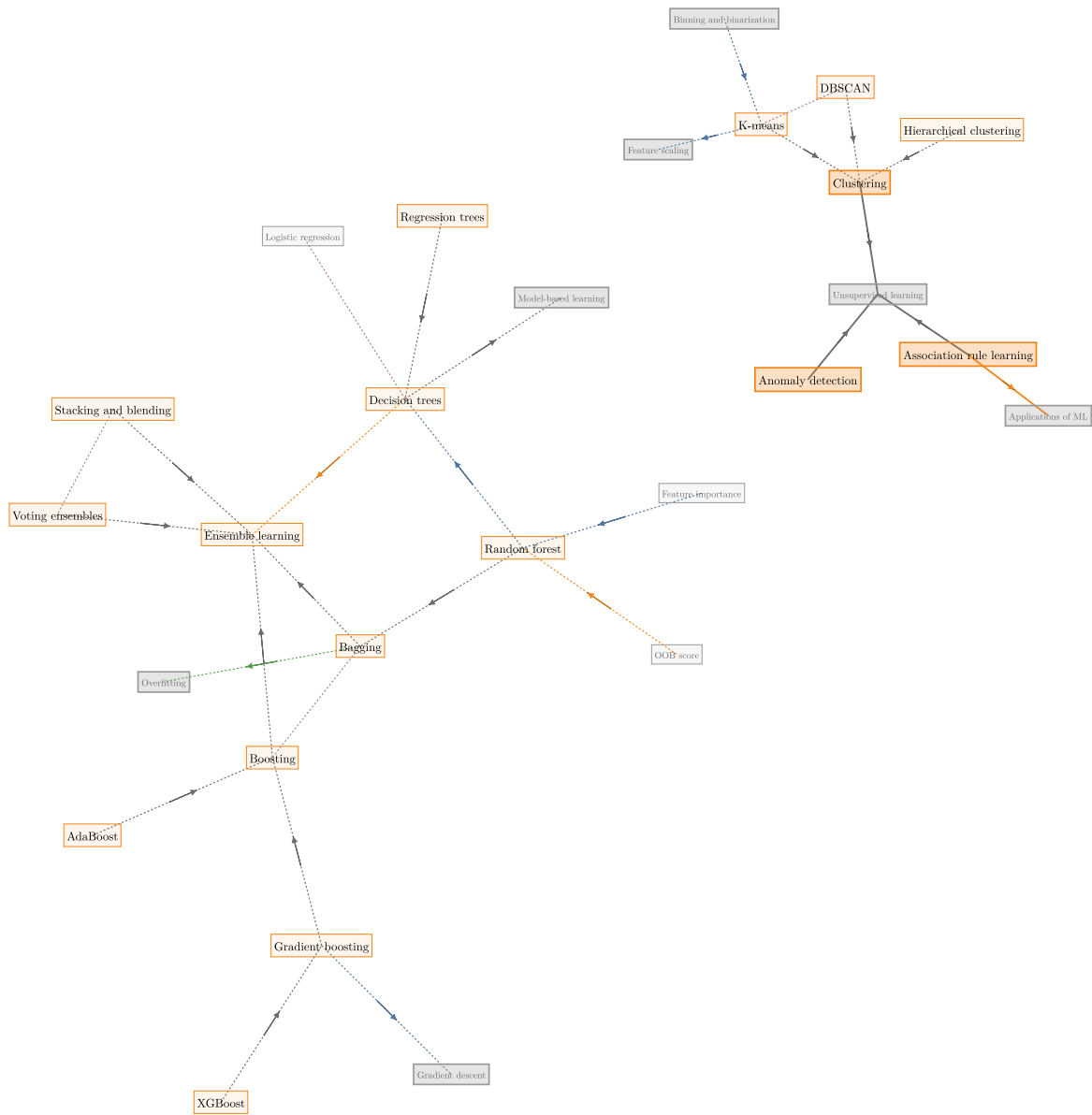


Figure 6: Concept map: Models: trees, ensembles and clustering

4. The Learning path

Each row lists a Video's Concepts and the Notes to read first. Videos marked *coming* or *deferred* do not have a Note yet.

15

Video	Concepts	Read first	Note
2	Artificial intelligence, Deep learning, Features, Machine learning, Neural networks, Symbolic AI and expert systems	nothing	written
3	Anomaly detection, Association rule learning, Classification problems, Clustering, Dimensionality reduction, Labelled data, PCA, Regression problems, Reinforcement learning, Semi-supervised learning, Supervised learning, Unsupervised learning	Note 2	written
4	Batch (offline) learning, Deployment, Model drift, Retraining	nothing	written
5	Online learning, Out-of-core learning, Retraining, Stochastic gradient descent	Note 4	written
6	Feature scaling, Instance-based learning, K-nearest neighbours, Model-based learning	Note 3	written

Video	Concepts	Read first	Note
7	APIs, Deployment, Enough data, Feature construction and splitting, Feature engineering, Labelled data, MLOps and cost, Missing values, Outliers, Overfitting, Poor-quality data, Sampling noise and bias, Software integration, Underfitting, Web scraping	Note 2	written
8	Applications of ML, Data mining	Note 2 , Note 3	written
9	APIs, Beta and A/B testing, Bivariate and multivariate analysis, CSV files, Deployment, Ensemble learning, Exploratory data analysis, Feature construction and splitting, Feature engineering, Feature scaling, Feature selection, Framing an ML problem, Hyperparameter tuning, Imbalanced data, ML development life cycle, MLOps and cost, Missing values, Model drift, Outliers, Poor-quality data, Retraining, Saving models with pickle, Standardization, Univariate analysis, Web scraping	Note 3 , Note 4 , Note 5 , Note 7	written

Video	Concepts	Read first	Note
11	Features, One-hot encoding, Tensors	nothing	written
12	Setup: conda, Jupyter and Colab	nothing	written
13	Accuracy, CSV files, Data leakage, Deployment, Exploratory data analysis, Feature scaling, Feature selection, Logistic regression, ML development life cycle, ML pipelines, Saving models with pickle, Standardization, Train-test split	Note 3 , Note 6 , Note 7 , Note 9	written
14	Framing an ML problem	Note 3 , Note 4 , Note 5	written
15	CSV files	nothing	written
16	JSON and SQL data	nothing	written
17	APIs	Note 16	written
18	Web scraping	nothing	written
19	Correlation, Descriptive statistics, Exploratory data analysis, Variance	Note 15	written
20	Outliers, Skewness, Univariate analysis	Note 9 , Note 19	written
21	Bivariate and multivariate analysis, Correlation	Note 19 , Note 20	written
22	Pandas Profiling	Note 12	written

Video	Concepts	Read first	Note
23	Binning and binarization, Encoding categorical data, Feature construction and splitting, Feature engineering, Feature extraction, Feature scaling, Feature selection, Feature transformation, Missing values, One-hot encoding, Outliers, Simple imputation (mean, median, mode, constant)	Note 11 , Note 13 , Note 20 , Note 21	written
24	Feature scaling, Standardization	Note 13 , Note 19 , Note 23	written
25	Normalization	Note 24	written
26	Encoding categorical data, Ordinal and label encoding	Note 13 , Note 23	written
27	Multicollinearity, One-hot encoding	Note 26	written
28	Column transformer	Note 23 , Note 26	written
29	Cross-validation, Deployment, Grid and random search, ML pipelines, Saving models with pickle	Note 13 , Note 17 , Note 23 , Note 28	written
30	Cross-validation, Function transformer, Q-Q plot	Note 20 , Note 23	written
31	Power transformer	Note 20 , Note 23 , Note 30	written
32	Binning and binarization	Note 23	written
33	Mixed variables	Note 23	written
34	Date and time features	Note 15 , Note 23	written

Video	Concepts	Read first	Note
35	Complete case analysis, Missing values	Note 9 , Note 20	written
36	Missing values, Simple imputation (mean, median, mode, constant)	Note 9 , Note 13	written
37	Missing values, Simple imputation (mean, median, mode, constant)	Note 9 , Note 13	written
38	Grid and random search, ML pipelines, Missing indicator, Missing values, Random sample imputation	Note 28 , Note 29 , Note 30 , Note 37	written
39	KNN imputer	Note 6 , Note 38	written
40	Iterative imputation (MICE)	Note 37 , Note 38	written
41	Capping (winsorization), IQR outlier method, Outliers, Percentile outlier method, Trimming outliers, Z-score outlier method	Note 9 , Note 19 , Note 20 , Note 24	written
42	Capping (winsorization), Trimming outliers, Z-score outlier method	Note 19 , Note 20 , Note 24 , Note 41	written
43	Capping (winsorization), IQR outlier method, Trimming outliers	Note 19 , Note 20 , Note 41	written
44	Capping (winsorization), Percentile outlier method, Trimming outliers	Note 41	written

Video	Concepts	Read first	Note
45	Feature construction and splitting	Note 23 , Note 30 , Note 32	written
46	Curse of dimensionality, Dimensionality reduction, Feature extraction, Feature selection	Note 3 , Note 21 , Note 23	written
47	Feature extraction, PCA, Variance	Note 19 , Note 23 , Note 24 , Note 46	written
48	Covariance and covariance matrix, Eigenvectors and eigenvalues, PCA	Note 3 , Note 24 , Note 46 , Note 47	written
49	PCA	Note 24 , Note 46 , Note 47 , Note 48	written
50	Best-fit line and squared error, Simple linear regression	Note 3 , Note 6 , Note 27	written
51	Best-fit line and squared error, Ordinary least squares (closed form), Simple linear regression	Note 3 , Note 6 , Note 27 , Note 48	written
52	Regression metrics	Note 51	written
53	Multiple linear regression	Note 3 , Note 51	written
54	Multiple linear regression, Normal equation	Note 3 , Note 51	written
55	Multiple linear regression, Normal equation	Note 3 , Note 51	written
56	Assumptions of linear regression	Note 27 , Note 30	written
57	Convex and non-convex loss, Gradient descent, Learning rate	Note 24 , Note 51	written
58	Batch gradient descent	Note 57	written

Video	Concepts	Read first	Note
59	Stochastic gradient descent	Note 57	written
60	Mini-batch gradient descent	Note 57	written
61	Overfitting, Polynomial features, Polynomial regression, Underfitting	Note 13 , Note 55	written
62	Bias-variance trade-off	Note 61	written
63	Regularisation, Ridge regression	Note 55 , Note 57 , Note 61 , Note 62	written
64	Ridge regression	Note 55 , Note 57 , Note 62 , Note 63	written
65	Ridge regression	Note 55 , Note 57 , Note 62 , Note 63	written
66	Ridge regression	Note 55 , Note 57 , Note 62 , Note 63	written
67	Lasso regression	Note 63	written
68	Lasso regression	Note 63	written
69	Elastic Net	Note 27 , Note 63 , Note 66 , Note 68	written
70	Logistic regression, Perceptron trick	Note 3 , Note 6 , Note 57 , Note 61	written
71	Logistic regression, Perceptron trick	Note 3 , Note 6 , Note 57 , Note 61	written
72	Logistic regression, Sigmoid function	Note 6 , Note 57 , Note 61 , Note 71	written
73	Log loss (binary cross entropy), Logistic regression	Note 57 , Note 61 , Note 71 , Note 72	written
74	Sigmoid function	nothing	written
75	Logistic regression	Note 61 , Note 71 , Note 73 , Note 74	written
76	Accuracy, Confusion matrix	Note 9 , Note 13	written
77	Precision, recall and F1	Note 9 , Note 76	written
78	ROC curve and AUC	Note 76 , Note 77	written
79	Softmax regression	Note 27 , Note 75	written
80	Polynomial features	nothing	written

Video	Concepts	Read first	Note
81	Hyperparameter tuning	nothing	written
82	Conditional probability	nothing	written
83	Independent and mutually exclusive events	nothing	written
84	Independent and mutually exclusive events	nothing	written
85	Bayes' theorem	Note 82	written
86	Bayes' theorem	Note 82	written
87	Naive Bayes	Note 3 , Note 84 , Note 86	written
88	Naive Bayes	Note 3 , Note 84 , Note 86	written
89	Naive Bayes	Note 3 , Note 84 , Note 86	written
90	Naive Bayes	Note 3 , Note 84 , Note 86	written
91	Accuracy, Cross-validation, Curse of dimensionality, Data leakage, Decision surface and boundary, Hyperparameter tuning, K-nearest neighbours, Overfitting, Underfitting	Note 3 , Note 6 , Note 13 , Note 24	written
92	Support vector machines	Note 3	coming
93	Support vector machines	Note 3	coming
94	Support vector machines	Note 3	coming
95	Kernel trick	nothing	coming
96	Kernel trick	nothing	coming
97	Decision trees	Note 6	coming

Video	Concepts	Read first	Note
98	Decision trees, Hyperparameter tuning	Note 6	coming
99	Regression trees	Video 98, coming	coming
100	Decision trees	Note 6	coming
101	Ensemble learning	Video 100, coming	coming
102	Voting ensembles	Video 101, coming	coming
103	Voting ensembles	Video 101, coming	coming
104	Voting ensembles	Video 101, coming	coming
105	Bagging	Note 91 , Video 101, coming	coming
106	Bagging	Note 91 , Video 101, coming	coming
107	Bagging	Note 91 , Video 101, coming	coming
108	Random forest	Video 100, coming, Video 107, coming	coming
109	Bias-variance trade-off, Random forest	Note 61 , Note 91 , Video 100, coming, Video 107, coming	coming
110	Random forest	Video 100, coming, Video 107, coming	coming
111	Hyperparameter tuning, Random forest	Video 100, coming, Video 107, coming	coming
112	Cross-validation, Grid and random search	Note 37 , Note 38 , Video 111, coming	coming
113	OOB score	nothing	coming
114	Feature importance	Video 111, coming	coming
115	AdaBoost	nothing	coming
116	AdaBoost	nothing	coming
117	AdaBoost	nothing	coming
118	AdaBoost, Hyperparameter tuning	nothing	coming
119	Boosting	Video 101, coming	coming
120	Gradient boosting	Note 57 , Video 119, coming	coming
121	Gradient boosting	Note 57 , Video 119, coming	coming
122	Gradient boosting	Note 57 , Video 119, coming	coming

Video	Concepts	Read first	Note
123	XGBoost	Video 122, coming	coming
124	XGBoost	Video 122, coming	coming
125	XGBoost	Video 122, coming	coming
126	XGBoost	Video 122, coming	coming
127	Stacking and blending	Video 101, coming	coming
128	Clustering, K-means	Note 3 , Note 24	coming
129	K-means	Note 24 , Video 128, coming	coming
130	K-means	Note 24 , Video 128, coming	coming
131	Hierarchical clustering	Video 128, coming	coming
132	DBSCAN	Video 128, coming	coming
133	Imbalanced data	nothing	coming
134	Optuna	Video 118, coming	coming

5. The Algorithm chooser

Key point: Two questions narrow the choice: do we have labelled outputs, and are they numbers or categories?

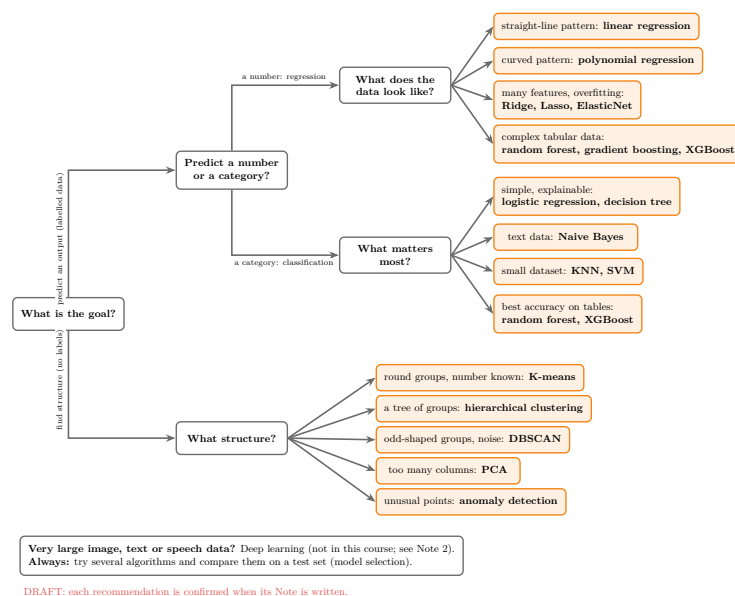


Figure 8: The Algorithm chooser (draft)

Figure 8 is a starting point, not a rule: in practice we try several suitable algorithms and

compare them on a test set. It is a draft until the algorithm Notes are written.

6. Key terms

Term	Meaning
Course map	The overview that ties all Notes together, in four views
Pipeline map	Where each Concept sits among the 14 steps of an ML project
Concept map	How Concepts are connected by Links
Learning path	Which Notes to read before which
Algorithm chooser	A flowchart from a problem's properties to suitable algorithms
Concept	One idea on the Course map
Link	A labelled connection between two Concepts
Draft / confirmed	Guessed from titles / checked against a written Note